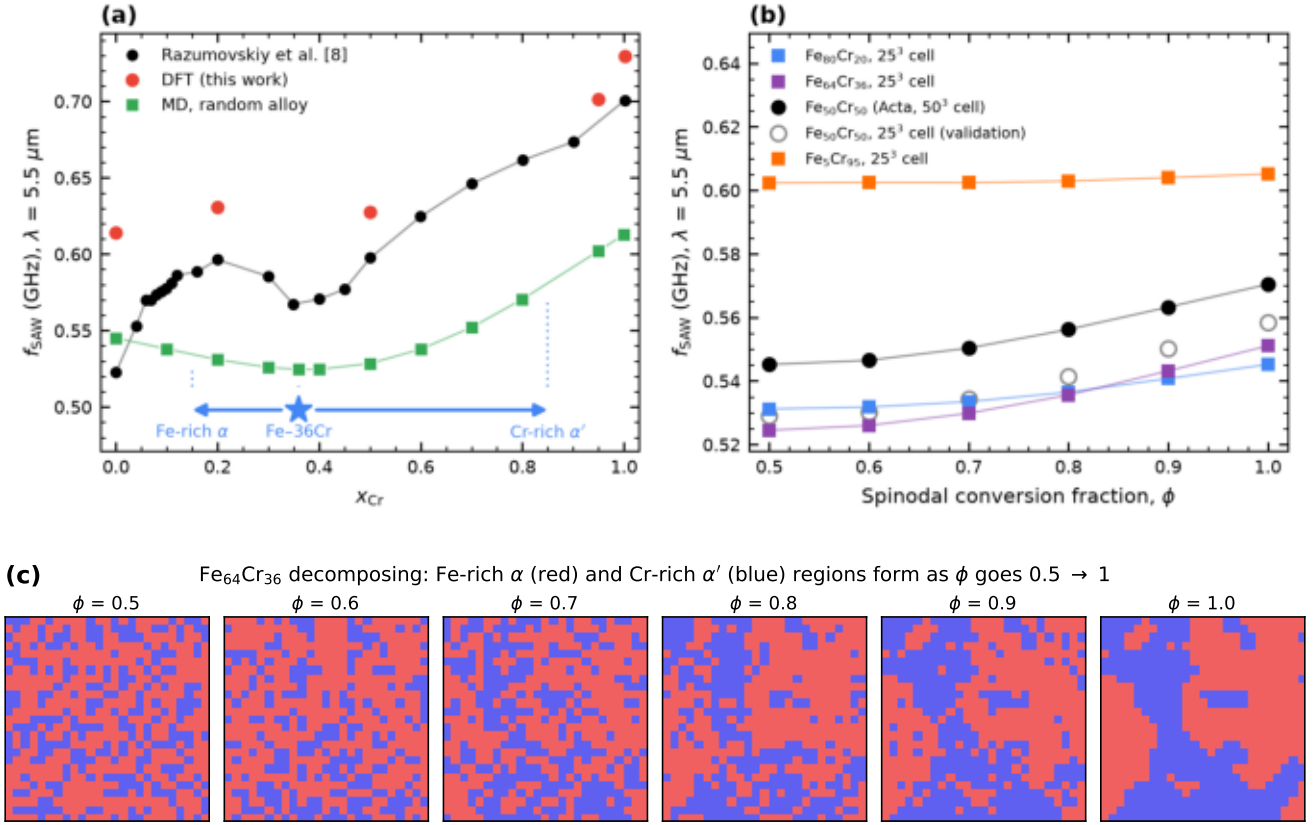

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
$f_{SAW}$ (GHz), $\lambda$ = 5.5 $\mu$m
$x_{Cr}$
Fe-rich $\alpha$
Fe-36Cr
Cr-rich $\alpha'$
(b)
$Fe_{80}Cr_{20}$, $25^3$ cell
$Fe_{64}Cr_{36}$, $25^3$ cell
$Fe_{50}Cr_{50}$ (Acta, $50^3$ cell)
$Fe_{50}Cr_{50}$, $25^3$ cell (validation)
$Fe_5Cr_{95}$, $25^3$ cell
$f_{SAW}$ (GHz), $\lambda$ = 5.5 $\mu$m
Spinodal conversion fraction, $\phi$
(c)
$Fe_{64}Cr_{36}$ decomposing: Fe-rich $\alpha$ (red) and Cr-rich $\alpha'$ (blue) regions form as $\phi$ goes 0.5 → 1
$\phi$ = 0.5
$\phi$ = 0.6
$\phi$ = 0.7
$\phi$ = 0.8
$\phi$ = 0.9
$\phi$ = 1.0
one atomic (001) plane of a $25^3$ bcc MD cell: red = Fe, blue = Cr